MUD SUSTAINABLE HOMES



CONSTRUCTION AND SAFETY REPORT

PT MUD SUSTAINABLE HOMES CONSTRUCTION AND SAFETY REPORT

22ND SEPTEMBER 2026

FROM: PT MUD SUSTAINABLE HOMES
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1. Introduction

This Construction and Safety Report has been prepared for Maine Residence, Unit E7-B, Nusa Dua, following concerns regarding the construction quality, deterioration, safety, hygiene, and suitability of the residence for continued occupation. The residence is approximately one year old, yet a number of significant issues have already become evident, including early material deterioration, the beginnings of black mould, concerns with moisture and water management, and the reported presence of E. coli in the property's water system. Other construction and safety deficiencies have also been identified that may affect the health, safety, and general wellbeing of the occupants.

The inspection and assessment have been undertaken from the perspective of 35 years of professional experience in construction, engineering, architecture, and related fields. I have operated construction companies in both Australia and Indonesia. In Australia, I am an Unlimited Domestic Licensed Builder, and in Indonesia I have been the Head Director of the above listed construction and architecture company since 2016.

The purpose of this report is to document the significant construction, safety, health, and maintenance issues identified at the residence and to provide a professional assessment of their condition and potential implications for the occupants. The report is intended to provide a clear technical record of the property's present condition and to assist the occupants and relevant parties in assessing whether the residence is suitable for continued residential occupation.

2. External Condition of the Residence

2.1 Observed Conditions

During the inspection of the external areas of the residence, the following conditions were observed:

- The external gate was in poor condition and was heavily dragging on the ground when operated.
- Cracking was observed in sections of the external brickwork and masonry.
- Several areas of external plasterwork showed significant cracking and deterioration.
- Above the main entry door, sections of plaster approximately 25–30 mm thick were observed to be heavily cracked.
- External wall paintwork was observed to be peeling, bubbling, discoloured and inconsistent in appearance, with areas showing signs of mildew and mould.
- Bubbling of the paintwork was observed in multiple locations.
- Early signs consistent with black mould growth were visible in some external areas.
- An exposed drainage sump/pit was observed. The design and condition of the opening appeared susceptible to the accumulation of debris and blockage.
- The occupants reported having observed rats entering the property from the vicinity of the drainage sump.
- Exposed electrical wiring associated with external lighting was observed.
- Numerous areas of chipped, damaged and poorly finished external brickwork were observed.
- The external paint application appeared visually inconsistent, including areas that appeared to have inadequate coverage or insufficient coating thickness.

2.2 Professional Assessment

Based on my 35 years of experience in construction, engineering, architecture and related fields, the condition observed externally represents premature deterioration for a residence approximately one year old.

The cracking and apparent instability of the heavier plaster sections above the entry door are of particular concern. Based on their location, thickness and observed cracking, these sections should be considered a potential falling-object hazard and should be professionally assessed and made secure. If a section were to detach and fall onto a person, the potential for significant injury exists.

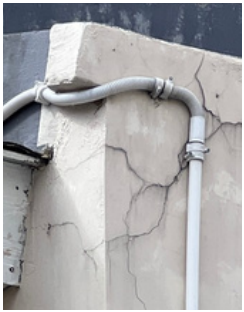
The extensive bubbling and peeling of external paint is consistent with moisture being present beneath the coating. In the climate of Bali, continued moisture exposure can contribute to deterioration of masonry and render and can provide conditions conducive to mould and mildew growth. The presence of visible mould at this stage is of particular concern given the substantially greater moisture exposure expected during the wet season.

The exposed electrical wiring represents a potential electrical safety hazard, particularly given the property's exposure to humidity, rain and external environmental conditions. The wiring should be appropriately enclosed, protected and assessed by a suitably qualified electrical professional.

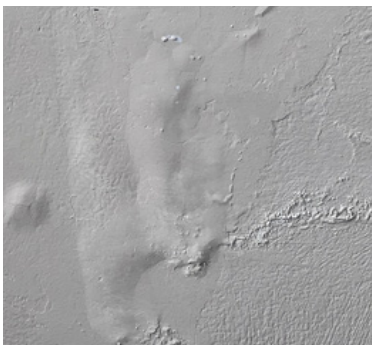
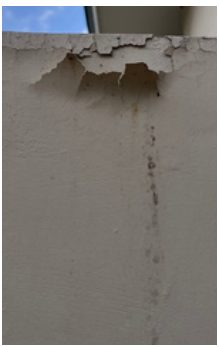
The exposed drainage sump also warrants attention. Its apparent susceptibility to blockage, together with the occupants' reported observations of rats entering from this location, raises drainage, hygiene and pest-control concerns.

Taken collectively, the observed external defects indicate significant deficiencies in workmanship, finishing, weather protection and maintenance for a building of approximately one year of age. The extent of deterioration observed warrants further investigation and appropriate rectification rather than being regarded as normal age-related wear.

BELOW I HAVE HIGHLIGHTED PHOTOS OF AFFECTED AREAS OF THE EXTERNAL STRUCTURE AND POOL/TERRACE AREA






3. Bedroom One – Condition and Ventilation

3.1 Observed Conditions

The following conditions were observed in Bedroom One:

- The bedroom has no conventional window.
- The only external opening is a poor-quality aluminium double sliding door providing access to the room.
- The residence as a whole appears to have no conventional windows, with similar aluminium sliding doors used as the primary openings.
- Significant cracking was observed in the bedroom masonry, including both vertical and horizontal cracks.
- Cracking was also observed extending vertically from several electrical power points.
- The walls show evidence of repeated patching or attempts to conceal previous defects, with visible differences in colour and finish.
- The external and internal wall finishes show areas of discolouration, fading and inconsistent paint coverage.
- There is no mechanical exhaust or dedicated ventilation system within the bedroom.
- The room experiences significant heat accumulation during the daytime, particularly in the absence of a conventional window or other effective ventilation opening.

3.2 Professional Assessment

The absence of a conventional window and the lack of mechanical ventilation significantly limit the room's ability to achieve natural ventilation and effective heat and moisture dissipation.
The bedroom faces the road and, in my professional opinion, would have been capable of incorporating a suitably designed window or larger ventilated opening. Such an opening could provide natural daylight and assist with the removal of accumulated heat and moisture.

The observed cracking around electrical outlets is also of concern. The pattern of cracking is consistent with the possibility that electrical conduits have been installed by chasing into the masonry after the blockwork was constructed, with insufficient depth or inadequate reinstatement of the plaster. This should be further investigated to establish the precise cause.

The combination of limited ventilation, heat accumulation, existing masonry cracking, moisture exposure and visible mould/mildew creates conditions that are not desirable for a healthy residential sleeping environment. These conditions are particularly relevant in Bali's climate, where humidity and moisture levels increase substantially during the wet season.

In my professional assessment, continued occupation of the room without addressing the ventilation, moisture and construction deficiencies is likely to result in ongoing deterioration of the finishes and an increased risk of mould and mildew development, particularly during periods of high humidity and rainfall.

The requirement to rely substantially on air-conditioning to control the room's temperature and environment, in the absence of adequate natural or mechanical ventilation, is also not considered an appropriate substitute for properly designed ventilation in a residential bedroom.
Overall, Bedroom One demonstrates significant deficiencies in ventilation, environmental control, masonry condition and finishing quality for a residence of approximately one year of age.

BELOW I HAVE HIGHLIGHTED PHOTOS OF AFFECTED AREAS IN BEDROOM 1.











4. Bathroom – Condition and Water Quality

4.1 Observed Conditions

The bathroom is one of the areas of the residence showing comparatively little visible deterioration.
The bathroom is designed as an open-air external bathroom, allowing substantial natural ventilation, daylight and air movement. Unlike the enclosed areas of the residence, the bathroom is able to dry naturally following use.

The occupants arranged for the property's water supply to be independently tested by the regional Health Laboratory of Bali (UPTD Balai Laboratorium Kesehatan Provinsi Bali). Testing was conducted across multiple sample points within the property, yielding the following Laboratory results:

- Kitchen water: Recorded a total coliform count of 727 CFU/100ml and 1 CFU/100ml of E. coli bacteria.
- Shower water: Recorded a total coliform count of 849 CFU/100ml and 1 CFU/100ml of E. coli bacteria.
- Shower filter: Recorded a total coliform count of 813 CFU/100ml and E. coli of <1 CFU/100ml (not detected).

The laboratory certificate and medical certificate of these tests, the tenants will combine with this construction and safety report.
As a construction professional, i cannot independently comment or give my opinion on the safety of the water supply to the villa.

The occupants have also reported that one of them subsequently became ill and was medically tested, with the medical testing reportedly returning a positive result for E. coli. Supporting medical and water test documentation provided by the occupants should be retained with this report.

4.2 Professional Assessment

The relatively good physical condition of the bathroom appears, in my professional assessment, to be assisted by its open-air design. Natural ventilation, daylight and continuous air movement allow moisture to dissipate more effectively and the bathroom to dry naturally. This is likely to reduce the conditions that contribute to mould and mildew growth.

Regarding the water system, as a construction professional i do not independently test water or provide medical evaluations, but i record the data provided by the occupants. The laboratory certificates document the total coliform counts between 727 and 849 CFU/100ml across the tested outlets and the presence of E. coli (1 CFU/100ml) in the sampled kitchen water, while WHO safe drinking guidance identifies the sbsence of E. coli in a 100ml sample as an important indicator of microbiological safety.

Although the occupants do not use the property's water system for drinking, the same water is used routinely for showering, bathing, washing, shaving and other personal hygiene activities, creating direct contact with the water supply.

Nevertheless, the documented presence of E. coli in water used for daily personal hygiene represents a water-quality and health concern. The source of contamination should be urgently investigated, the water system appropriately treated, and independent follow-up testing undertaken to confirm that the water is suitable for normal household and personal-hygiene use. This water-quality issue is an important consideration when assessing the overall suitability of the property for continued residential occupation.

5. Pool and Entry Areas

5.1 Observed Conditions

Pool

The following conditions were observed around the swimming pool:

- The pool water level was noticeably low at the time of inspection, indicating a potential leak or significant water loss.
- The occupants confirmed that the pool water level drops significantly on a daily basis.
- Significant deterioration of the grout between the pool coping tiles was observed, with numerous sections of grout loose, missing or appearing to have popped out.
- The condition of the coping and grout indicates that water may be able to penetrate beneath the coping tiles.

Entry Areas

Significant cracking and deterioration were observed in the plasterwork above the external entry doors, particularly above the main entrance leading into the kitchen area.

The affected plaster sections are relatively thick and appear to have poor adhesion or bonding to the underlying brickwork. Considerable cracking is already visible despite the relatively young age of the building.

The front door entry lock also seems to display a fault, the key only exiting the barrel if the door is open. when locking the door to secure the property the key becomes stuck as the lock has the inability to perform a full rotation for the key to be removed. This is of course a security issue as the tenants cannot lock the door to secure the property.

5.2 Professional Assessment

The ongoing loss of pool water should be investigated as a potential leak within the pool structure, plumbing or waterproofing system. A properly constructed pool should not experience substantial unexplained daily water loss beyond normal evaporation and reasonable operational losses.

The deteriorated coping grout is also of concern. Failed or missing grout allows water to penetrate beneath the coping system. Continued water penetration, particularly during the wet season, may contribute to further deterioration and loosening of the coping tiles.

This creates a potential slip and falling-object hazard. A coping tile that becomes loose while a person is standing on or stepping across the pool edge could move or detach unexpectedly, potentially causing a person to lose their balance or fall into the pool.

The heavily cracked plaster above the entry doors presents a more direct potential falling-material hazard. While plaster itself is generally a non-structural finish, the sections observed are substantial in thickness and are located directly above areas where occupants regularly pass.

In my professional assessment, the apparent lack of adequate bonding of these heavy plaster sections to the masonry requires urgent attention. External environmental exposure, moisture and vibration or movement of the building may contribute to further deterioration. In a region subject to seismic activity, unsecured heavy overhead finishes should not be disregarded as a potential safety issue.

If a substantial section of plaster were to detach while a person was standing below it, it could result in serious injury, including potentially significant head injury.
The pool and entry areas therefore demonstrate additional examples of premature deterioration and apparent deficiencies in workmanship and finishing, together with conditions that have potential safety implications for the occupants of the residence.

BELOW I HAVE HIGHLIGHTED PHOTOS OF AFFECTED AREAS IN THE POOL AND ENTRY AREAS.







6. Kitchen Area

6.1 Observed Conditions

The following conditions were observed or reported in the kitchen area:

- Internally, above the entry doors, approximately 30 mm of plaster appears to have separated from the underlying brickwork, with significant cracking visible between the plaster and masonry.
- Further cracking was observed throughout the kitchen walls, including vertical cracking extending from electrical outlets.
- Significant cracking was observed at the junction between the walls and ceiling.
- Previous repairs and patching were visible in several areas, together with inconsistent, faded and apparently inadequate paint coverage.
- Sections of grout were missing between floor or wall tiles around the entry area.
- The property's water system was tested at the kitchen supply. The laboratory certificate records 727 Coliform count and 1CFU/100ml water of E. coli bacteria from the kitchen tap. (Certificates provided by tenants)
- The gas installation beneath the stove was inspected visually. An older gas cylinder and associated tubing were present. The regulator appeared comparatively new; however, the tubing connections appeared to use inappropriate or inadequate clamping arrangements at the regulator and cylinder connections.
- The stove, tapware, sink and electrical outlets appear to be low-cost, basic-grade fixtures and fittings.
- The occupants reported that two electrical outlets have previously short-circuited.
- The occupants also reported frequent loud impacts occurring on the roof, apparently from objects falling onto the roof. The occupants provided video evidence of one such occurrence. They report that this has occurred regularly since the beginning of their tenancy.
- The source of the objects falling onto the roof could not be established during the inspection.

6.2 Professional Assessment

The heavily cracked plaster above the entry doors is of particular concern because of its location, thickness and apparent separation from the masonry. Although plaster is normally a non-structural finish, a substantial section that is inadequately bonded and positioned directly above a doorway represents a potential falling-material hazard.
Movement or vibration of the building, including movement associated with seismic activity, could potentially cause further separation. If a substantial section were to fall while an occupant was passing underneath, the potential for significant injury exists.

The gas installation also requires attention. The apparent use of inadequate clamps at the gas connections is a potential fire and gas-leak hazard, particularly because the installation is directly below the cooking appliance. This should be inspected and certified by a suitably qualified gas technician before continued use.

The missing grout around the tiled areas provides potential pathways for moisture to penetrate beneath the tile system. Continued moisture penetration can contribute to deterioration of adhesives, substrates and surrounding finishes and may eventually result in loose or displaced tiles.

The reported electrical short-circuiting incidents are concerning. The electrical wiring and conduits are predominantly concealed within the walls and could not be fully inspected during this visual assessment. Consequently, I cannot confirm the condition or standard of the concealed electrical installation without further investigation and testing.

However, given the reported electrical failures and the general standard of workmanship observed elsewhere in the property, I would recommend that the complete electrical installation be professionally inspected and tested, rather than assuming that the concealed system is compliant.

Similarly, only limited plumbing pipework was visually accessible during the inspection. The visible pool plumbing appeared to use Rucika-brand PVC pipework, which is a commonly available economical plumbing product. The presence of economical materials alone does not establish that an installation is defective; however, the limited visibility of the concealed plumbing means that its condition and installation standards cannot be verified during this inspection.

The repeated unexplained impacts on the roof are also concerning. The source could not be confirmed, and therefore no definitive conclusion can be made regarding their cause. Nevertheless, the occupants' report that these events have occurred regularly throughout their tenancy, together with the supplied video evidence, indicates an ongoing unresolved condition affecting the normal enjoyment and use of the residence. The source should be investigated to determine whether material is detaching from the building or whether another external cause is responsible.

Overall, the kitchen demonstrates further evidence of poor finishing, premature deterioration and potentially inadequate construction practices, together with specific concerns relating to the gas installation, electrical system, water quality and overhead plasterwork. Several of these issues have direct or potential implications for the health, safety and continued residential occupation of the property.

As previously stated above, i cannot offer my opinion or recommendations regarding the water supply quality or the tenants health issues.

BELOW I HAVE HIGHLIGHTED PHOTOS OF AFFECTED AREAS IN THE KITCHEN AREA.













7. Bedroom Two – Master Bedroom

7.1 Observed Conditions

Bedroom Two is located directly off the kitchen and is considered the master bedroom of the residence. The following conditions were observed:

- The bedroom has no external window and no dedicated mechanical ventilation to the outside.
- Significant evidence of moisture damage and mould growth was observed, particularly on the wall behind the bed.
- Cracking is present throughout the room in vertical, horizontal and diagonal directions.
- Cracking was also observed extending vertically from electrical outlets.
- Significant cracking and separation were observed at various wall-to-ceiling junctions.
- The opening made in the wall for installation of the air-conditioning unit appears to be substantially oversized, with the air-conditioning installation not completely sealing the opening.
- The unsealed area provides an opening through which insects and other pests may enter and potentially inhabit the wall or air-conditioning installation.
- The occupants report that the bedroom becomes extremely hot during the daytime, with substantial heat accumulation within the room.
- The wall behind the bed, which adjoins the neighbouring building, shows significant evidence of moisture penetration and deterioration.
- The paintwork throughout the room shows areas of deterioration, discolouration and inconsistent coverage.

7.2 Professional Assessment

The combination of no external window, no dedicated ventilation, significant heat accumulation and existing moisture and mould damage is of particular concern in a bedroom used for sleeping.

The wall behind the bed requires further investigation to determine the source and extent of the moisture penetration. As this wall adjoins the neighbouring building, the construction arrangement between the two buildings could not be confirmed by visual inspection. If a cavity or concealed space exists between the structures, moisture entering this area could potentially remain trapped and contribute to further deterioration. The condition of any concealed structural or masonry components cannot be confirmed without intrusive investigation.

The observed cracking throughout the room, including the wall-to-ceiling junctions and cracking extending from electrical outlets, provides further evidence of poor finishing and construction quality and should be professionally investigated to establish the underlying causes.

The oversized air-conditioning opening is also inadequately sealed. Apart from reducing the effectiveness of the air-conditioning system, the opening provides a potential pathway for insects, moisture and other environmental contaminants to enter the room or wall cavity.

The reported extreme heat within the bedroom is particularly relevant given the existing moisture and mould conditions. A warm, poorly ventilated environment with elevated humidity provides favourable conditions for the continued growth of mould and mildew. These conditions may become more pronounced during Bali's wet season when humidity and moisture levels increase.

In my professional assessment, the current condition of Bedroom Two represents a poor and potentially unhealthy sleeping environment. The combination of inadequate ventilation, excessive heat, moisture damage, visible mould, extensive cracking and defective finishing requires investigation and rectification.

If these conditions remain unresolved, there is a significant risk of continued deterioration of the room and worsening moisture and mould conditions, particularly during the wet season. The bedroom should therefore not be considered satisfactory for continued residential use without appropriate investigation and remedial works.

BELOW I HAVE HIGHLIGHTED PHOTOS OF AFFECTED AREAS IN BEDROOM, 2 AREA.











8. Conclusion – Health, Safety and Suitability for Occupation

Based on my inspection and assessment, the Maine Residence, Unit E7-B, Nusa Dua, demonstrates a level of construction deficiency, premature deterioration and environmental concern that is not consistent with what would reasonably be expected of a residence approximately one year old.

Throughout the residence, there are repeated indications of poor workmanship, inadequate finishing, moisture penetration, cracking masonry and plasterwork, insufficient ventilation, mould and mildew development, and deficiencies in building services. In addition, the property water system, tested from the kitchen supply, has been professionally tested and recorded, the tenants will supply the relevant certificates and reports along with this construction and safety report. However, the test results are a concern for occupants using this water for showering, bathing and personal hygiene.

Several conditions identified also have potential physical safety implications, including inadequately bonded and heavily cracked plaster above doorways, deteriorated pool coping grout, cloudy pool water, exposed electrical wiring, concerns regarding the gas installation, and reported electrical failures. While not every observed defect represents an immediate danger, the cumulative condition of the property presents a level of risk that warrants serious consideration.

The absence of adequate natural or mechanical ventilation in the enclosed bedrooms is particularly concerning when considered together with the existing moisture damage, mould growth and substantial heat accumulation. In Bali's climate, these conditions can become significantly more difficult to control during the wet season, when increased rainfall and humidity provide more favourable conditions for moisture retention and mould and mildew development.

It is also understood that the owners have repeatedly arranged for tradespeople to enter the residence to undertake patching, repairs and remedial works in response to continuing defects. While necessary maintenance is an ordinary part of property ownership, the frequency and continuing nature of these interventions, where related to recurring construction deficiencies, can have a significant impact on the occupants' privacy, quiet enjoyment and normal use of their home.

A residence should provide occupants with a reasonably safe, healthy and stable environment in which to live. Continual defects requiring repeated access by tradespeople, together with ongoing deterioration and unresolved health and safety concerns, substantially interfere with that normal residential environment.

In my professional opinion, the number, nature and recurrence of the issues identified indicate that the problems are not isolated cosmetic defects, but form a broader pattern of construction, ventilation, moisture-management, finishing and building-service deficiencies.

Unless these matters are comprehensively investigated and properly rectified, I consider there to be a continuing risk of further deterioration and increasing health and safety concerns for the occupants, particularly as the property enters the wet season.

The condition documented in this report therefore raises serious concerns regarding the ongoing suitability of the residence for safe and healthy residential occupation.

Kind Regards,
Greg Lloyd (Director Utama)
PT Mud Sustainable Homes